SALEEM IBRAHIM

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Education

Université de Montréal — Montréal, QC
M.Sc. Computer Science (Machine Learning), GPA: 4.0 — *Sept. 2025 – June. 2027*

Machine Learning Specialization — Online (Coursera)
Stanford University & DeepLearning.AI — *Jun. 2023 – Aug. 2023*

Wilfrid Laurier University — Waterloo, ON
Honours Bachelor of Science — Computer Science and Psychology (Double Major) — *Aug. 2018 – Jun. 2023*

Dean's Honor Roll: 2019/2020, 2021/2022, 2022/2023

Responsible AI Scholarship Recipient

Projects

SoccerNet Dense Video Captioning | *PyTorch, Transformers, Multimodal Learning, Contrastive Learning* 2026

- Improved the SoccerNet-Caption baseline for dense video captioning by replacing NetVLAD-based temporal aggregation with transformer-based video representations and multimodal audio-visual modeling.
- Investigated contrastive pretraining between soccer video features and frozen language embeddings, as well as modality-specific cross-attention for integrating visual and crowd-audio information.
- Designed and evaluated transformer and dual-LSTM decoder variants through controlled ablation studies, analyzing the contribution of audio, contrastive learning, and multimodal fusion to captioning performance.
- Improved CIDEr performance over the original SoccerNet baseline while reducing model size, demonstrating more effective representations for temporally grounded soccer caption generation. ‟

NHL Expected Goals (xG) Analytics Platform | *Scikit-learn, Flask, Docker, Streamlit* December 2025

- Built an end-to-end machine learning pipeline to predict shot quality in NHL games (xG), including data ingestion, cleaning, and feature engineering.
- Developed and evaluated Boosted Trees, Neural Networks, Logistic Regression, and SVM models on shot-context features (distance, angle, game state), leveraging ROC/AUC metrics and calibration analysis.
- Deployed the model as a Flask REST API in Docker and integrated a Streamlit dashboard for interactive analytics.

Diabetic Retinopathy Classification | *Python, PyTorch, Scikit-learn, CNNs, Transfer Learning* October 2025

- Built a 5-class retinal disease severity classifier using fundus images spanning No DR to Proliferative DR.
- Built ML pipelines for highly imbalanced datasets, leveraging data augmentation to significantly improve classification performance.
- Developed custom CNNs and transfer learning models (e.g., ResNet) analyzing overfitting and data limitations.
- Benchmarked deep learning approaches against classical models such as SVMs and Random Forests.

Experience

Research Intern — May 2026 – Present
WatchOut / Adonite — *Montréal, QC*

- Developing a multi-turn diagnostic system that asks targeted clarifying questions, reasons over competing hypotheses, and iteratively resolves ambiguous issues.
- Investigating Bayesian experimental design (BED) for adaptive question selection, using expected information gain (EIG) to quantify how informative candidate questions are for reducing uncertainty over hypotheses.
- Researching reinforcement-learning approaches for multi-turn information gathering, including GRPO-based optimization of question-asking policies under information-gain and task-outcome reward objectives.
- Working on designing controlled evaluation environments and diagnostic benchmarks to study information-seeking behavior, decision making, convergence toward correct hypotheses, and premature conclusions.

Product Analyst — Jan 2024 – May 2025
D2L

- Supported data-driven product improvements by reviewing functional and technical specifications, analyzing user behavior patterns in Kibana and Honeycomb, and submitting structured Feature Requests informed by recurring client issues and platform usage trends.
- Improved product reliability by identifying, reproducing, and documenting bugs with clear technical context (logs, traces, steps to reproduce, expected vs. actual behavior), accelerating engineering triage and resolution cycles.
- Built and maintained SQL queries and diagnostic scripts to investigate production issues, analyze client data, and surface actionable insights for complex case resolution.
- Resolved high-volume technical support requests under strict SLAs by correlating customer-reported issues with system logs, metrics, and traces, translating technical root causes into clear client-facing solutions.

Teaching and Learning Systems Assistant May 2021 – Aug 2022
Wilfrid Laurier University *Remote*

- Provided front-line technical support for Laurier's Brightspace LMS, troubleshooting system issues and escalating complex cases to engineering and IT teams.
- Developed testing scenarios for enterprise system upgrades and maintained XML-based email templates supporting teaching workflows.
- Migrated course content, grades, and assessments into Brightspace while assisting instructors with test and assessment setup.

Technical Skills

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Jupyter
Data Engineering & Storage: SQL, MongoDB
Programming Languages: Python, Java, C/C++, JavaScript — Familiar: Rust, Prolog, Kotlin
MLOps & Cloud: Docker, Git, Linux/Unix, AWS, Google Cloud, Kibana, Elasticsearch, honeycomb
Web & Backend: React, Next.js, Django, Firebase

Leadership & Extracurriculars

Mila Lab Representative 2025 – Present
Mila - Quebec AI Institute *Montreal, QC*

- Elected student representative supporting research community engagement, internal initiatives, and communication between students and Mila administration.
- Coordinated events, feedback collection, and operational projects improving student experience across research teams.

Responsible AI Scholarship Recipient 2025
RAI group *Toronto, ON*

- Selected for conference scholarship recognizing academic excellence and commitment to ethical, socially responsible artificial intelligence.

Neuro-AI Reading Group & Reinforcement Learning Reading Group
Mila - Quebec AI Institute *Montréal, QC*

- Participating in weekly paper discussions on biologically inspired learning, representation learning, and modern reinforcement learning methods.